Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	09 January 2026
Team ID	
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Image Upload System	The system shall allow users to upload microscopic blood cell images in formats such as JPG, PNG, or JPEG for malaria analysis.
FR-2	Image Preprocessing	The system shall preprocess uploaded images by resizing, normalizing, and preparing them for input into the deep learning model.
FR-3	Malaria Detection Model	The system shall use trained deep learning models (Custom CNN, MobileNetV2, EfficientNetB0) to classify the uploaded image as Parasitized or Uninfected.
FR-4	Result Display	The system shall display the prediction result to the user through the web interface along with the uploaded image.
FR-5	Model Integration with Web Application	The system shall integrate the trained model with a Flask backend to process user requests and return predictions.
FR-6	User Interaction Interface	The system shall provide a web-based interface built with HTML, CSS, and JavaScript for easy interaction and image upload.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system shall provide a simple and user-friendly web interface that allows users to easily upload blood cell images and view prediction results without requiring technical knowledge.
NFR-2	**Security**	The system shall ensure secure handling of uploaded images and prevent unauthorized access to the system and stored data.
NFR-3	**Reliability**	The system shall consistently provide accurate predictions and maintain stable operation during multiple user requests.
NFR-4	**Performance**	The system shall process uploaded images and return prediction results within a few seconds to ensure real-time analysis.
NFR-5	**Availability**	The system shall remain available and accessible to users whenever the web application is running on the server.
NFR-6	**Scalability**	The system shall be capable of handling an increased number of users or image uploads by upgrading computational resources or deploying the application on cloud platforms.